

AI Handbook Assistant using RAG

Submitted By

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Track B — RAG & Knowledge Systems

Introduction

The AI Handbook Assistant using RAG is a beginner-level AI project developed using Python in Google Colab.

This project helps users retrieve answers from handbook data using a simple Retrieval Augmented Generation (RAG) concept.

The system searches relevant handbook information and displays answers based on user questions entered by the user.

The main purpose of this project is to understand:

- Basic Artificial Intelligence concepts
 - Prompt Engineering
 - Retrieval-based response systems
 - Python programming
 - Knowledge retrieval from documents
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Objective

- To create a simple AI assistant using Python
 - To understand the basic concept of RAG
 - To retrieve answers from handbook data
 - To implement prompt engineering techniques
 - To build a beginner-friendly AI application
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Problem Statement

Students often face difficulty finding information from large handbook documents manually.

This project solves that problem by creating a simple AI assistant that retrieves relevant information quickly from stored handbook data.

Technologies Used

Technology	Purpose
Python	Main Programming Language
Google Colab	Development Platform
Text File	Knowledge Base Storage
Prompt Engineering AI Instruction Design	

Project Working

The project works in the following steps:

1. Handbook data is stored inside a text file.
2. Python reads the handbook data.
3. User enters a question.
4. The system searches for matching information.
5. The relevant answer is displayed to the user.

This project demonstrates the basic idea of Retrieval Augmented Generation (RAG).

System Prompt

The following system prompt is used in the project:

```
SYSTEM_PROMPT = """
```

```
You are an academic handbook assistant.
```

Rules:

1. Answer only from handbook data.
2. Give short and simple answers.
3. If answer is not found, say:
'The handbook does not contain this information.'

""""

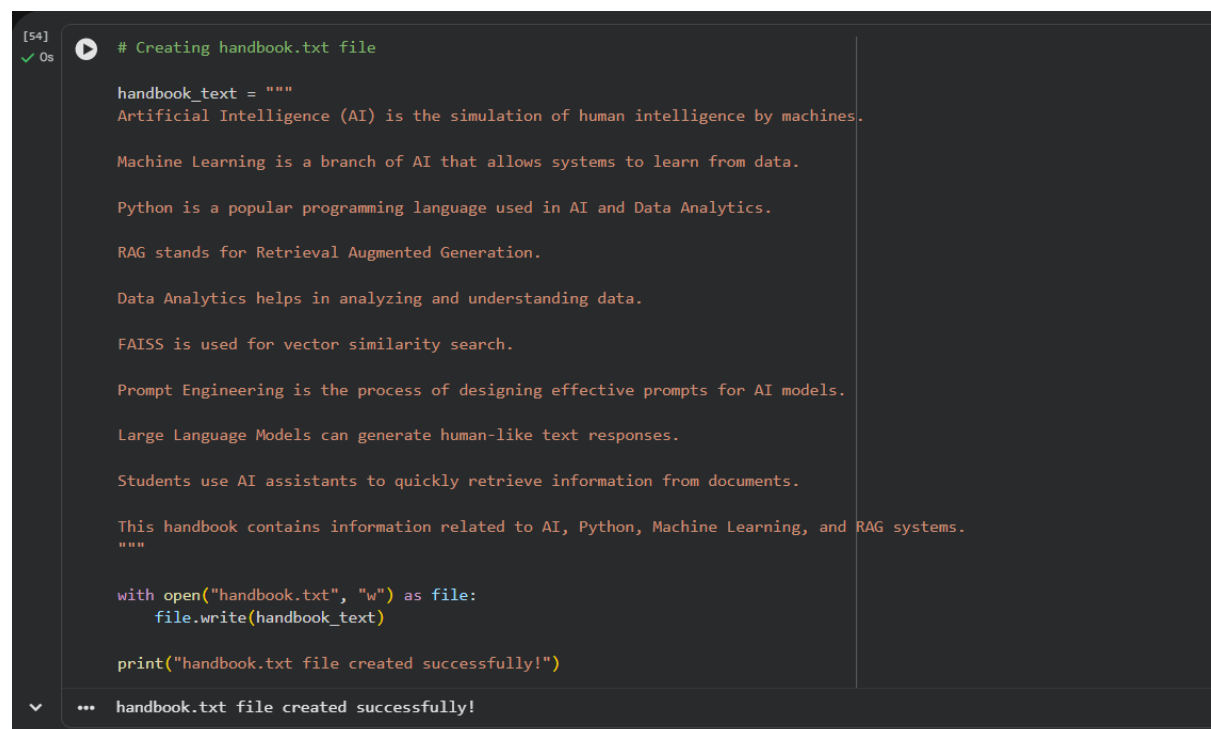
Code Implementation

The project was implemented using Python in Google Colab.

The following tasks were performed:

- Creating handbook data
- Reading handbook file
- Taking user input
- Searching relevant information
- Displaying answers

Screenshot 1 — Handbook Data File



The screenshot shows a Google Colab interface with a code cell. The code cell is titled "[54] ✓ 0s" and contains a Python script to create a text file named "handbook.txt". The script defines a multi-line string containing information about AI, Machine Learning, Python, RAG, Data Analytics, FAISS, Prompt Engineering, and Large Language Models. It then uses the "open" function to write this string to the "handbook.txt" file and prints a success message. The output of the code cell is visible at the bottom, showing "handbook.txt file created successfully!".

```
[54] ✓ 0s # Creating handbook.txt file

handbook_text = """
Artificial Intelligence (AI) is the simulation of human intelligence by machines.

Machine Learning is a branch of AI that allows systems to learn from data.

Python is a popular programming language used in AI and Data Analytics.

RAG stands for Retrieval Augmented Generation.

Data Analytics helps in analyzing and understanding data.

FAISS is used for vector similarity search.

Prompt Engineering is the process of designing effective prompts for AI models.

Large Language Models can generate human-like text responses.

Students use AI assistants to quickly retrieve information from documents.

This handbook contains information related to AI, Python, Machine Learning, and RAG systems.
"""

with open("handbook.txt", "w") as file:
    file.write(handbook_text)

print("handbook.txt file created successfully!")

... handbook.txt file created successfully!
```

Screenshot 2 — System Prompt Code

```
[55]
✓ 0s
SYSTEM_PROMPT = """
You are an academic handbook assistant.

Rules:
1. Answer only from handbook data.
2. Give short and simple answers.
3. If answer is not found, say:
    'The handbook does not contain this information.'
"""

print(SYSTEM_PROMPT)
```

▼

```
You are an academic handbook assistant.

Rules:
1. Answer only from handbook data.
2. Give short and simple answers.
3. If answer is not found, say:
    'The handbook does not contain this information.'
```

Screenshot 3 — Main Python Code

```
[56]
✓ 0s
# Load handbook data

with open("handbook.txt", "r") as file:
    data = file.readlines()

print("Handbook Loaded Successfully!")
```

▼

```
Handbook Loaded Successfully!
```

Screenshot 4 — Running the Program

```
[57]
✓ 3m
▶ print("=====")
print(" AI Handbook Assistant using RAG ")
print("=====")

while True:

    question = input("\nAsk your question (type exit to quit): ")

    if question.lower() == "exit":
        print("\nThank You!")
        break

    found = False

    print("\nSearching handbook...\n")

    for line in data:

        if question.lower() in line.lower():

            print("Answer:")
            print(line)

            found = True

    if found == False:
        print("The handbook does not contain this information.")
```

Screenshot 5 — AI Query Output

```
▼ ... =====  
AI Handbook Assistant using RAG  
=====  
  
Ask your question (type exit to quit): AI  
  
Searching handbook...  
  
Answer:  
Artificial Intelligence (AI) is the simulation of human intelligence by machines.  
  
Answer:  
Machine Learning is a branch of AI that allows systems to learn from data.  
  
Answer:  
Python is a popular programming language used in AI and Data Analytics.  
  
Answer:  
FAISS is used for vector similarity search.  
  
Answer:  
Prompt Engineering is the process of designing effective prompts for AI models.  
  
Answer:  
Students use AI assistants to quickly retrieve information from documents.  
  
Answer:  
This handbook contains information related to AI, Python, Machine Learning, and RAG systems.
```

Screenshot 6 — Python Query Output

```
Ask your question (type exit to quit): Python  
  
Searching handbook...  
  
Answer:  
Python is a popular programming language used in AI and Data Analytics.  
  
Answer:  
This handbook contains information related to AI, Python, Machine Learning, and RAG systems.
```

Screenshot 7 — RAG Query Output

```
Ask your question (type exit to quit): RAG  
  
Searching handbook...  
  
Answer:  
RAG stands for Retrieval Augmented Generation.  
  
Answer:  
This handbook contains information related to AI, Python, Machine Learning, and RAG systems.
```

Screenshot 8 — Exit Command Output

```
Ask your question (type exit to quit): exit  
  
Thank You!
```

Evaluation Test Cases

Query	Expected Result	Actual Result	Status
AI	AI Definition	Correct Answer	Pass
Python	Python Information	Correct Answer	Pass
RAG	RAG Meaning	Correct Answer	Pass
IPL Score	Refusal	Correct Refusal	Pass
Bitcoin	Refusal	Correct Refusal	Pass

Prompt Iteration

Prompt Version 1

Answer the user question from handbook.

Problem

- Sometimes unclear responses were generated.
-

Prompt Version 2

You are an academic handbook assistant.

Rules:

1. Answer only from handbook data.
2. If answer is missing, say:
"The handbook does not contain this information."

Improvement

- Improved accuracy
 - Reduced hallucination
 - Better grounded answers
-

Faithfulness Test

Question	Response	Hallucination?
Who won IPL 2025?	The handbook does not contain this information.	No
What is Bitcoin?	The handbook does not contain this information.	No
Current weather?	The handbook does not contain this information.	No

Advantages

- Beginner-friendly project
 - Easy to understand
 - Demonstrates basic AI concepts
 - Uses prompt engineering
 - Fast information retrieval
-

Limitations

- Works only on stored handbook data
 - Limited search capability
 - No advanced AI model integration
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Future Improvements

- Add graphical user interface (GUI)
 - Use real LLM APIs
 - Add PDF upload support
 - Improve retrieval accuracy
 - Add voice assistant feature
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Conclusion

The AI Handbook Assistant using RAG is a simple beginner-level AI project developed using Python and Google Colab.

The project demonstrates:

- Prompt Engineering
- Retrieval Augmented Generation (RAG)
- Knowledge Retrieval
- AI-based Question Answering

This project helped in understanding how AI systems retrieve information from stored knowledge sources and generate grounded responses.

The project successfully achieved its objectives and provided a practical understanding of beginner-level AI applications.